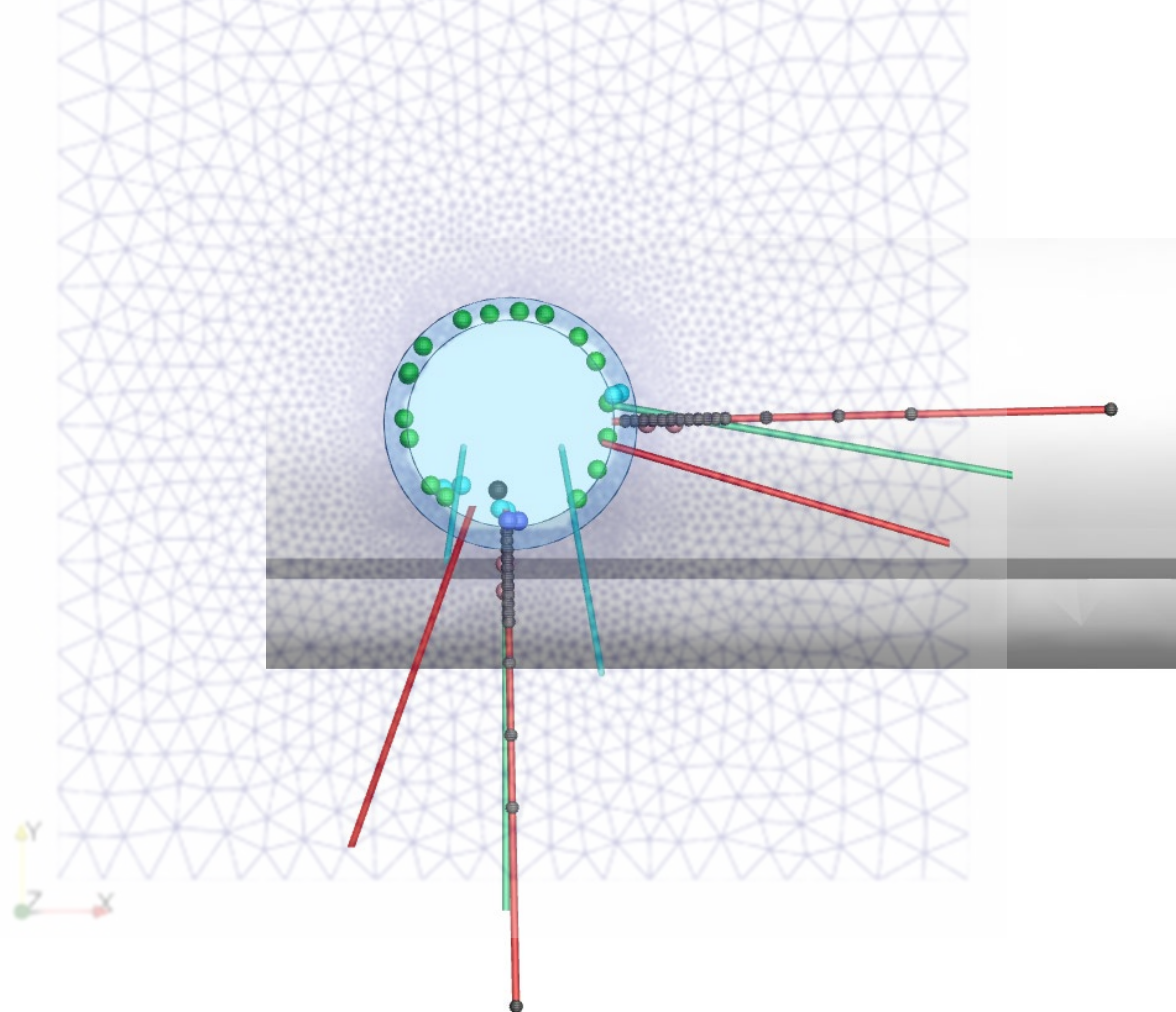


TODO



Difference related to $t_0=10/2019$, computed as given in publication

If a saturation field from OGS is available, which parameters need to be fitted in order to simulate the discrete measurements?

Permeability,
pressure BC at surface (derived from RH measurements and Kelvin equation),
retention curve (pc-S-relation),
(krel-S-relation, porosity)
(storativity),
BC (stress)

Would it be better to include the saturation-resistivity (*Archie, WC-Sat*) relation directly in the model (meaning computing resistivity between two points from OGS saturation), with its parameters treated as unknowns (*Archies parameter (2)*), so that less artificial error is introduced?

How reliable are the measurements, and what types of errors can occur (e.g. noise, fracture-related electric paths not represented in the OGS model)? *To be checked with Markus*

Given the instability observed at some physical points over time, will it be necessary to “clean” the measurements (e.g. discard some observations)? Have you performed such filtering before, or do you have guidance on how to distinguish erroneous from naturally expected variations? *To be checked with Markus*

The computational domains used for ERT fields and OGS simulations differ—how should this discrepancy be addressed? *to be discussed (slide)*



CD-A twin niches in the Mont Terri Rock Laboratory: Characterization and interpretation of hydraulic parameters with regard to safety aspects

²Ingenieurgesellschaft für Umwelt in der Umwelttechnik mbH (IEU), Karlsruhe, Germany

ARTICLE INFO

- Expendable
- Most Tecti Project
- Characterization of epulis in chondroste
- CD-A experiment
- Griped hydraulic mechanical effects
- Ontology
- Geological measurement

ABSTR

The State Tech High Laboratory conducted a comprehensive interdisciplinary exploration program that carried out to increase knowledge about long-term coupled hydrologic-mechanical effects in Opales Creek. As these effects are of significant interest regarding the stability and integrity of a potential seepage facility for high level waste, the "Behavior of Opales Creek" was chosen as the primary focus of the program. The program was characterized by Opales Creek, especially by the homogeneous sandy silt, and to investigate the evolution of the water content, pressure and deformation in the formation. This article provides an overview of measurement methods and presents the results of the investigation. The Opales Creek experiment was carried out in the CUBA experiment, which is located in the upper sandy silt of the Opales Clay which is characterized by the existence of different subclay types and fault traces. This coordination focuses on the detailed characterization of the Opales Creek, while the evolution of the water content, pressure and deformation is investigated during the first two years after the excavation of the siltbox have been carried out. They indicate that the

selection type and pre-existing fault zones do not affect the distribution of water content severely, which could be an essential finding in the context of detection procedures for potential high level waste (HLW) repositories. Nevertheless, a further analysis on wetting potential seems indicative, clearly, for the assessment of the EED and the permeability in the clayey zone. The clayey zone is the most critical zone gathered. After 1.5 years the EED and the permeability is much more pronounced in the open than in the closed beds. Future work aims at an increased process understanding based on a verified constitutive model for Opalinus Clay and a comprehensive system understanding considering long-term effects due to climatic conditions as well as

The geodetical measurements were performed using the high resolution earth resistivity meter 4-pole light 10 W (Lippmann Geophysikalische Messtechnik), controlled by the software GeotEC (Geophysik-Dr. Rausch). Within the CD-A experiment, high resolution borehole measurements along exploratory boreholes are used to

across the rock. This work was carried out soon shortly after the electrode is completed using the HPL pump (DRI-10 HPL-31). It consists of 100 electrodes with a spacing of 15 mm. The total has a diameter of 80 mm (the typical tube diameter of 80 mm), but greater diameter can be adapted. The electrode can be extended progressively for the required connection to the borehole wall. The spatial resolution is limited by 2 mm (electrode distance (30 mm), the maximum depth of investigation is about 20 mm perpendicular to the borehole wall. The chosen electrode configuration represents a Wenner or wing (Fig. 4).

A numerical inversion process is necessary for data analysis and interpretation. The measured apparent resistivity along the borehole wall is transformed into a spatial model, distributed into a depth

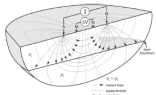


Fig. 5. Schwarz representation of a surface measurement is four-point measurement over a broad half space (after Ref. 43).

Seismic velocity measurements make use of elastic waves excited at defined source locations. The seismic wave field is recorded in terms of travel time and amplitude at certain receiver positions. Taking into account the acquisition geometry and travel time, propagation velocity of direct (primary (p-)) waves v_p and secondary shear (s-) waves v_s can be calculated. Typical velocity values for the relevant lithology obtained by in-situ borehole measurements at the Mont Terri laboratory are

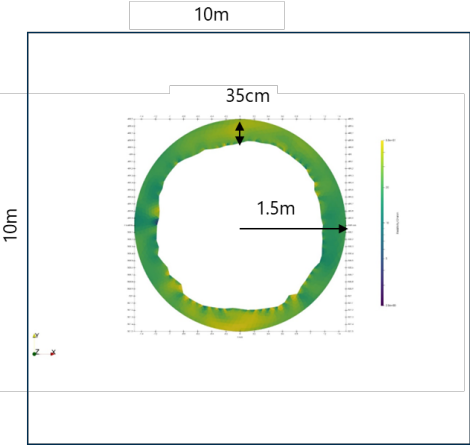
values around 2500 m/s, for the carbonate rich areas: values between 3000 and 4000 m/s, and for the sandy facies values around 3000 to 2500 m/s. In fault zones an average decrease of 200 to 400 m/s relative to the base value was observed. Near excavation the measured values for the intact and damaged zones are varying, reflected by the highly variable seismic signal with fluctuations from 1500 m/s up to 2600 m/s, as discussed by Ref. 22.

used to deduce a velocity model along travel paths of the waves in two (2D) or three spatial dimensions (3D). If repeat survey data are available, temporal data can be included and velocity changes determined (4D), to permit geological process analysis. For the acquisition of interval velocity data, special interval velocity measurement (PIM) are useful too. In the CDA experiment, a seismic borehole was drilled to a depth of 100 m and seven receivers are moved regularly along the surveyed boreholes with 5 m intervals. At each step, the probe armator and sensors are pneumatically attached to the borehole wall to establish sufficient mechanical coupling for the generation and detection of the emitted inquisitive ("shot") elastic waves (Fig. 6). The wave signals are recorded with seven receiver channels and one source channel (trigger channel).

4.3. Data sets and evaluation measures

Information about the permeability of the host rock is gained via pulse tests in the boreholes BPG-D19 to BPG-A27 and with evaporation measurements in the boreholes BPG-D19, W124, the boreholes BPG-A2 and BPG-A26 are aligned in vertical direction, boreholes BPG-D19 and BPG-A27 are aligned in horizontal direction and BPG-D19 exactly in horizontal direction as presented in Fig. 3. The two twin bores are located perpendicular to the strike of bedding with an inclination of 55°66' to the direction of the niches (145°53') which is summarized in Ref. 30 (detail). The pulse tests have been carried out in March 2021, 1.5 years after excavation of the niches, while the evaporation measurements have been carried out in summer (June/July 2020) about 10 months

In the pulse test, a modified COMREL double piston pump system (<http://www.comrel.de>) with a static interval length of 2 s is used. A non-toxic nitrogen gas at standard temperature and pressure was used as injection medium with an injection pressure up to 1 bar. The pulse test is characterized by the injection of the gas as a benchmarker monitoring of the pressure evolution within the test interval. This allows the determination of the intrinsic permeability of a 3 m volume in the near field around a 10 cm long borehole interval.

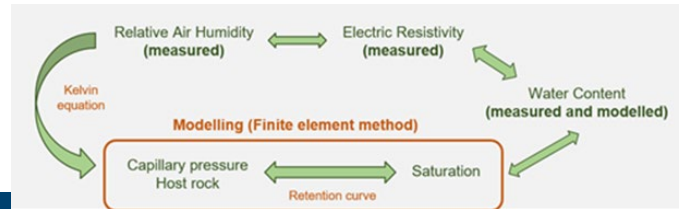
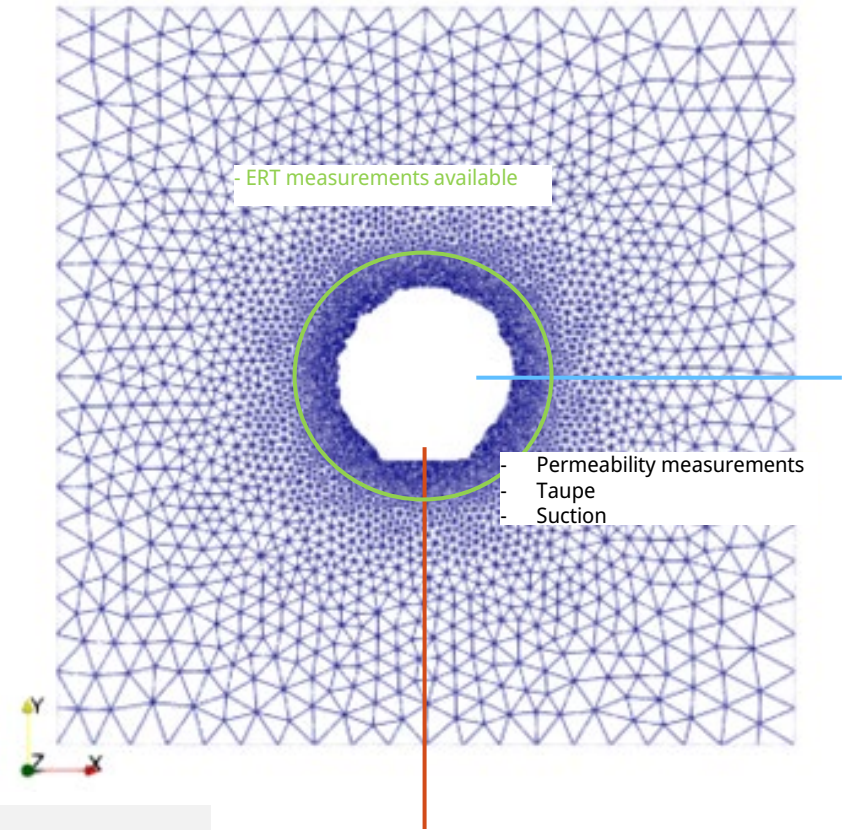


NMR measurements

- Would you be against including NMR measurements in the inverse problem? *No*
- If they are available, could they be incorporated into the Bayesian inverse formulation as additional measurements? *Yes*
- How reliable are these measurements? *To be checked with Stephan*

Inversion setup

- How should the permeability subdomains be specified (number, dimensions, etc.)? Will permeability be treated as scalar within subdomains? *To be discussed*
- How should storativity be treated? Since it is often linked to permeability through the porosity field, would it be better to consider porosity as an unknown and use parameterized relations for permeability and storativity? *To be discussed (we are in the desaturated zone...)*
- Which other parameters need to be specified? *Pressure BC at surface, pc-S-curve*

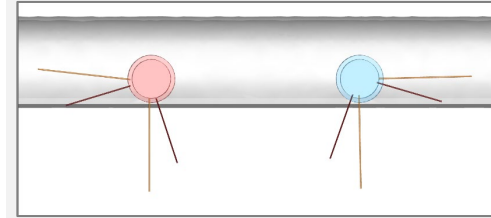
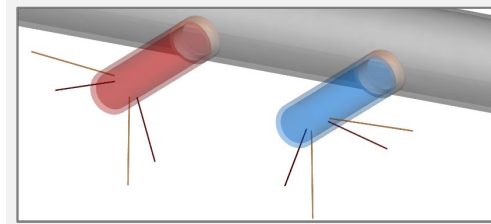
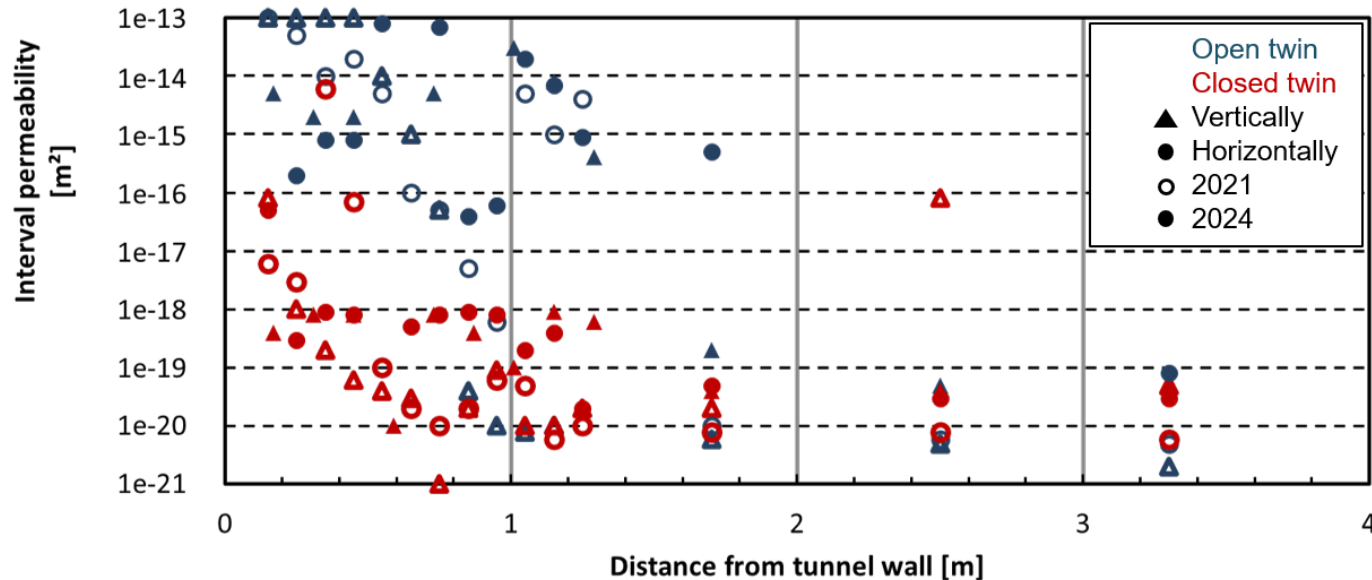


If we start with permeability only, I will need to confirm the other OGS input parameters (or verify if the values in the model files you provided are correct).

For permeability, I'll need your help defining the domains: for example, a ring around the niche with constant values on segments — the width and segmentation.

Permeability

Permeability (Pulse tests)



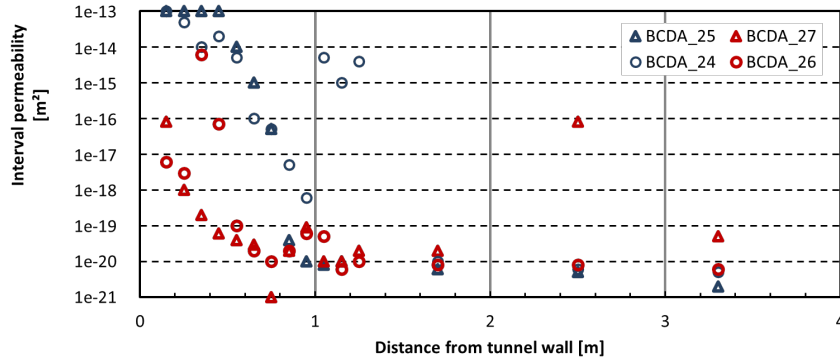
Measurements in 2021
(1.5 yrs after excavation)
Measurements in 2024
(4.5 yrs after excavation)

2 measurements in each
direction at every 10cm
interval (3 yrs inbetween)

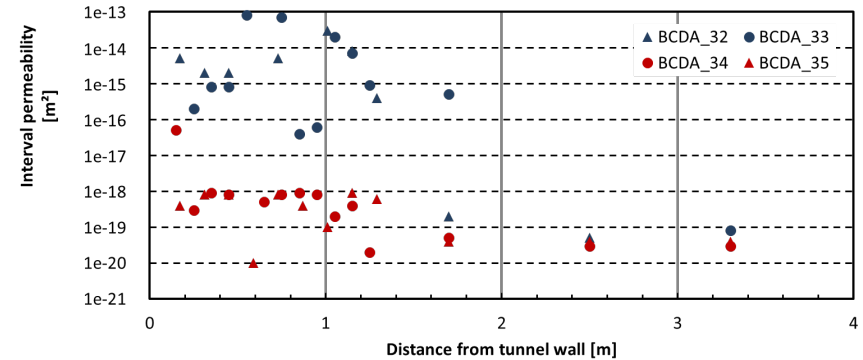
- modified COMDRILL double piston packer system (<http://www.comdrill.de/>)
 - static interval length of 10 cm
- non-reactive nitrogen gas at standard temperature and pressure was used as injection medium
 - injection pressure up to 1 bar
- injection of the gas and hereafter a monitoring of the pressure evolution within the test interval
- determination of the intrinsic permeability of a 3D volume in the near field around a 10 cm long borehole interval
 - maximum error from experimental and evaluation sources is half the order of magnitude

Permeability (Pulse tests)

2021 (1.5 yrs after installation)



2024 (4.5 yrs after installation)

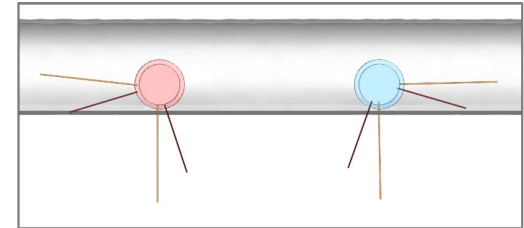


2021 as input in early stage
2024 later?

Rings:

One ring per measurement? As the curve (2021) is very steep...
But data points in 2024 are varying...

Open twin
Closed twin
▲ Vertically
● Horizontally
○ 2021
● 2024

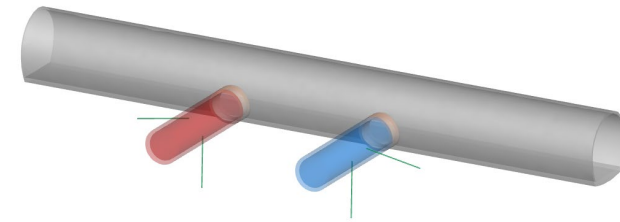
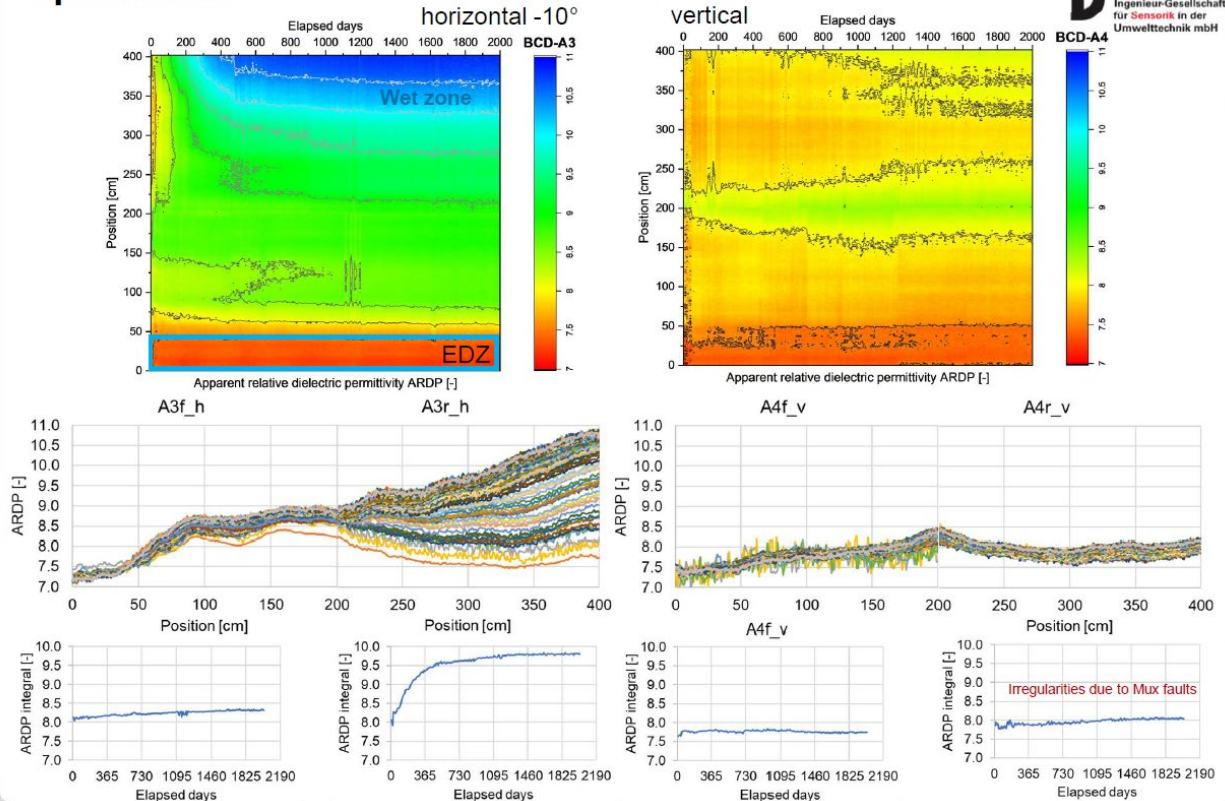


I think any measurement that has a reasonably clear connection to OGS model outputs could be valuable for the inverse problem. But I would need your help to understand what each measurement represents and how it links to model outputs. For example, I am not familiar with Taupe measurements (is that refractometry?). If you could provide a short overview of how **suction, permeability, or Taupe data relate** to OGS variables, that would help me a lot if we decide to use them.

Measurements related to saturation (Taupe, Suction, ERT, NMR)

Taupe (TDR measurements)

Open Twin



$$\varepsilon_r = \varepsilon_{\text{rock}} * (1 - \varphi) + \varepsilon_w * \varphi * S_w$$

With:

ε_r : residual apparent relative dielectric permittivity,
 $\varepsilon_{\text{rock}}$: apparent relative dielectric constant of rock,
 φ pores in the rock,
 ε_w : apparent relative dielectric constant of water,
 S_w : water saturation.

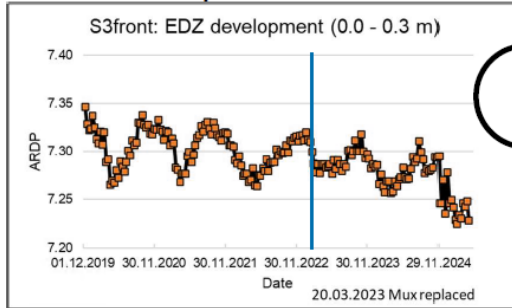
$$\Delta \varepsilon_r = \varepsilon_{\text{Gestein}} * (1 - \varphi) + \varepsilon_w * \varphi * \Delta S_w$$

ε_w (permittivity): ca. 80
 φ = Porosity * 100 [Vol.%pores]

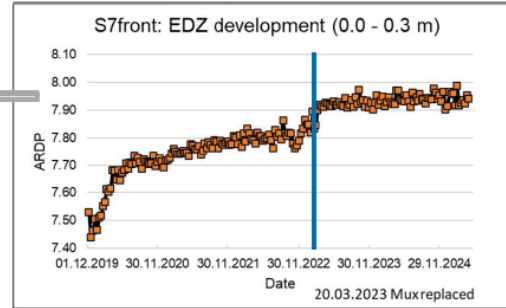
Meteorological influences on EDZ

(Mean amplitude over first 0.30 m)

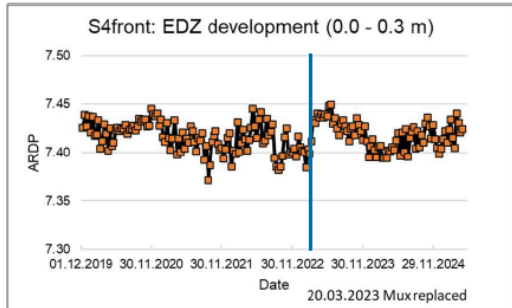
Open Twin



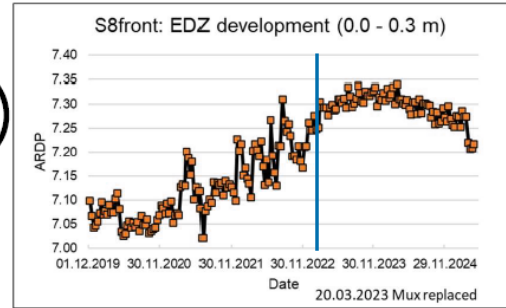
Closed Twin



Variability higher in horizontal direction

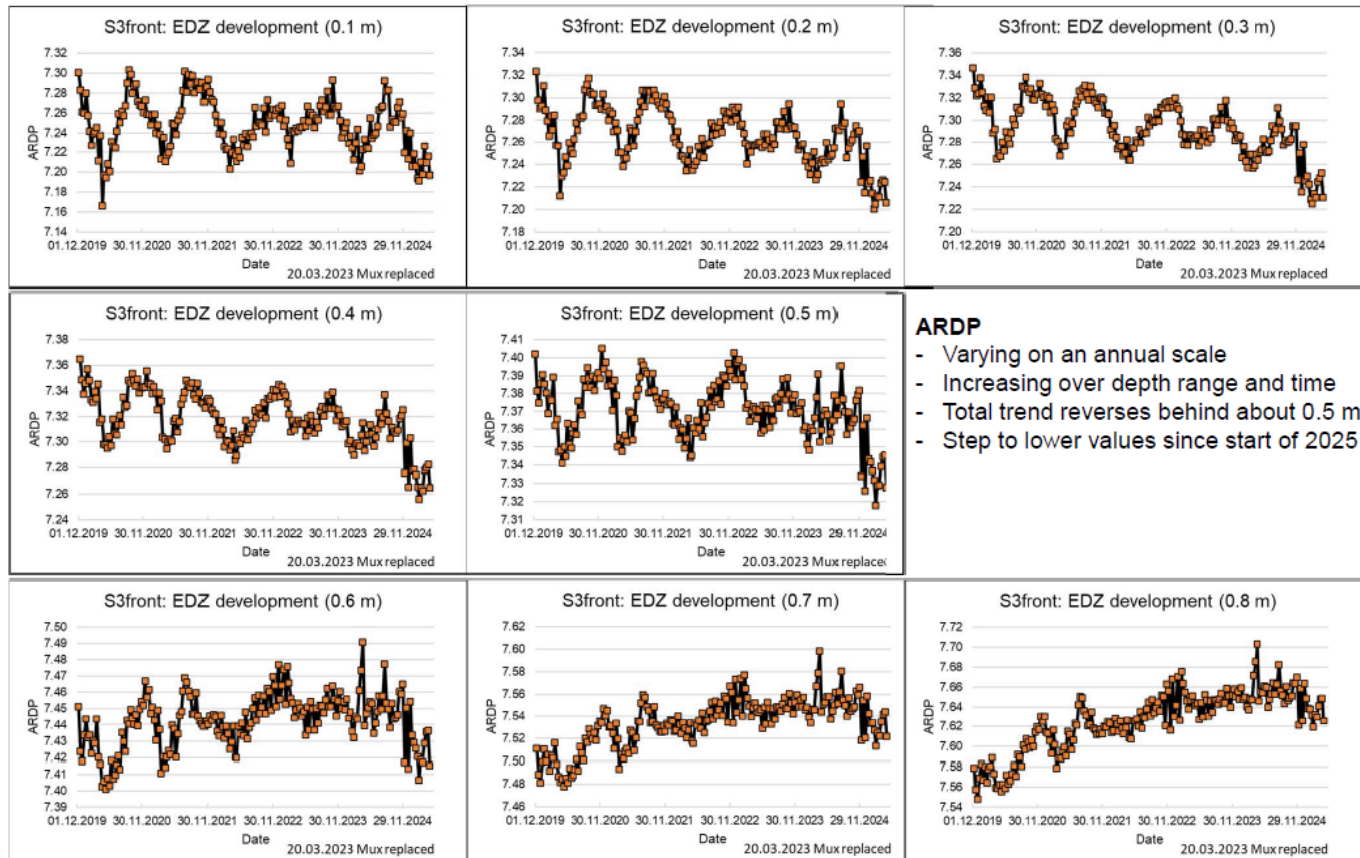


No visible influence in both directions



Meteorological influences on EDZ at S3f

(Mean amplitude over varying depth, starting at 0 m)



ARDP

- Varying on an annual scale
- Increasing over depth range and time
- Total trend reverses behind about 0.5 m
- Step to lower values since start of 2025

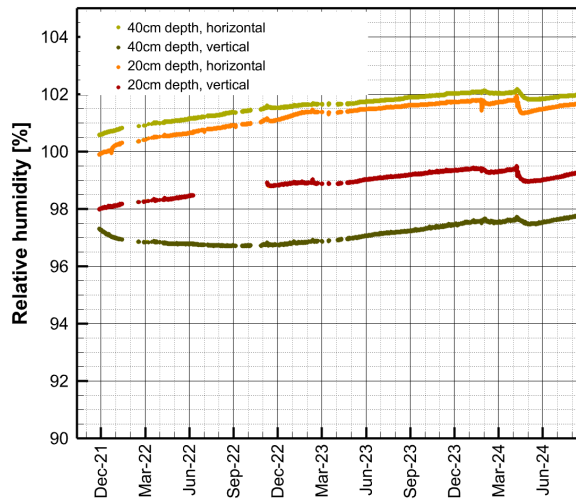
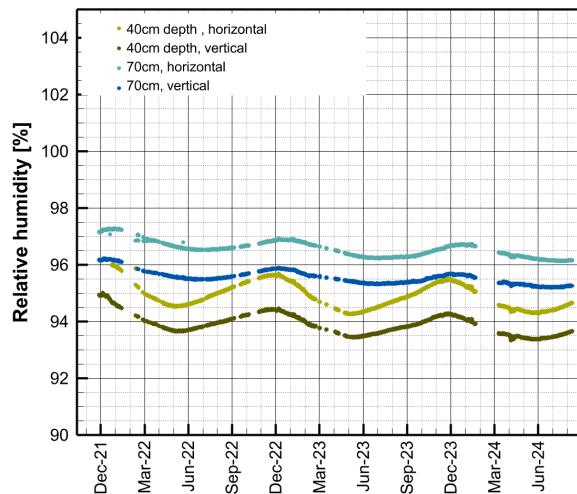
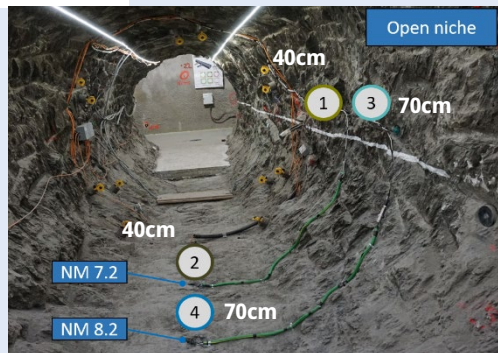
We can provide:
Mean amplitude of
change of saturation
S for different
intervals from the
niche wall

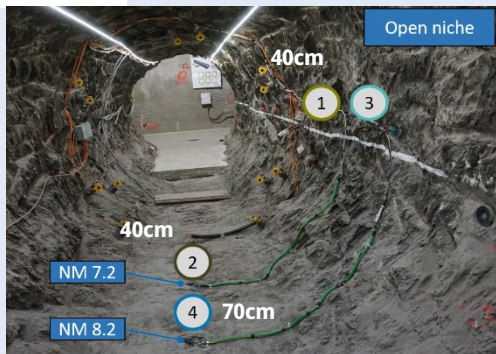
Intervals?
Proposal:
0-10cm
10-20cm
20-30cm

...

(different to plot,
where we have:
0-10cm
0-20cm
0-30cm
...)

Water content evolution at a depth of 40cm





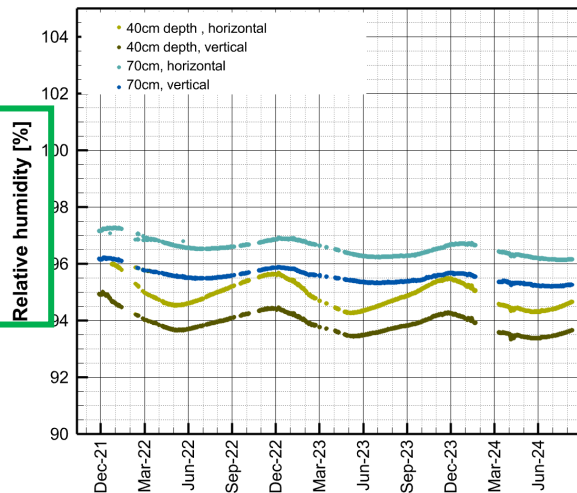
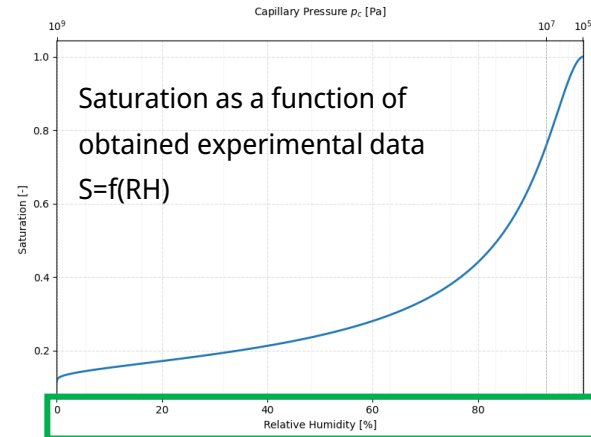
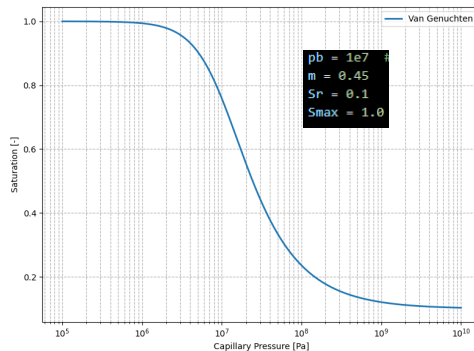
$$\frac{RH}{100} = \exp(p_L M) \rho R T$$

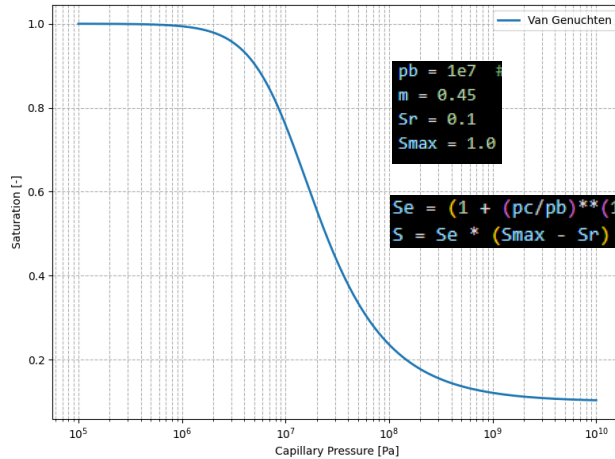
$$p_c = -p_L = \frac{\rho R T}{M} \ln\left(\frac{RH}{100}\right)$$

$$S_e = \left(1 + \left(\frac{p_c}{p_b}\right)^{\frac{1}{1-m}}\right)^{-m}$$

$$S = S_e(S_{max} - S_r) + S_r$$

Van Genuchten curve





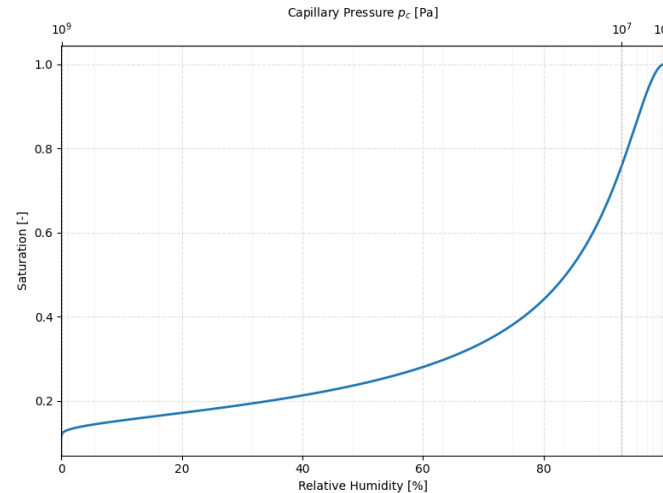
$$S_e = \left(1 + \left(\frac{p_c}{p_b}\right)^{\frac{1}{1-m}}\right)^{-m}$$

$$S = S_e (S_{max} - S_r) + S_r$$

```

RH_val = np.exp((pc * M) / (rho_L * R * T)) * 100
ln_term = np.log(RH_clamped / 100.0)
pc = (rho_L * R * T / M) * ln_term

```



$$p_L < 0:$$

$$\frac{RH}{100} = \exp(p_L M) \rho R T$$

$$p_L = -p_c = \frac{\rho R T}{M} \ln\left(\frac{RH}{100}\right)$$

$$S_e = \left(1 + \left(\frac{p_c}{p_b}\right)^{\frac{1}{1-m}}\right)^{-m}$$

$$S = S_e (S_{max} - S_r) + S_r$$

I know the NMR dataset is very sparse compared to ERT, but since it measures water content more directly than ERT plus Archie, it could still serve to anchor the inversion. I'd also like to compare the NMR values to the ERT-derived water content to better understand where the apparent noise in the Archie fit is coming from. The Rsquared looked rather low, I'm not sure if that's due to homogenization of ERT (I hope I got it right from the paper) or something else. Could you please share the NMR measurement tables (I assume for the open twin), along with the sensor positions, so I can link them with the ERT spatial fields?

NMR:
Stephan Costabel (BGR)
back on September 4th

Available data: NMR

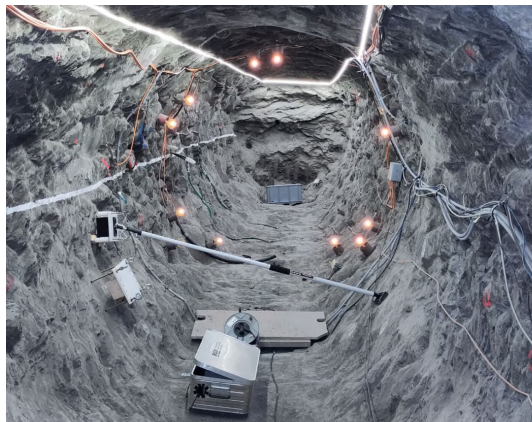
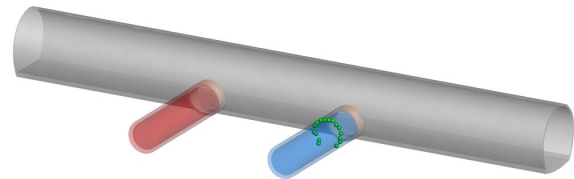
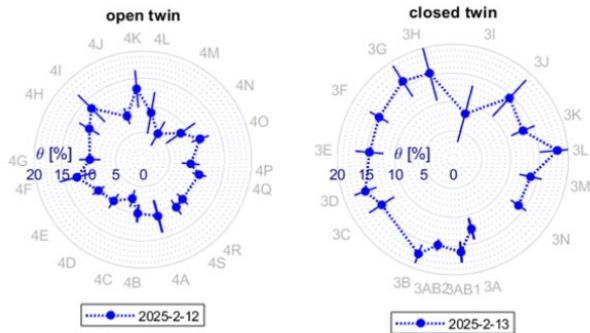
Data set 1: Water content at pre-defined positions

- 16/19 positions along profile
- Twice the year
- Adaptation of measuring system and post-processing

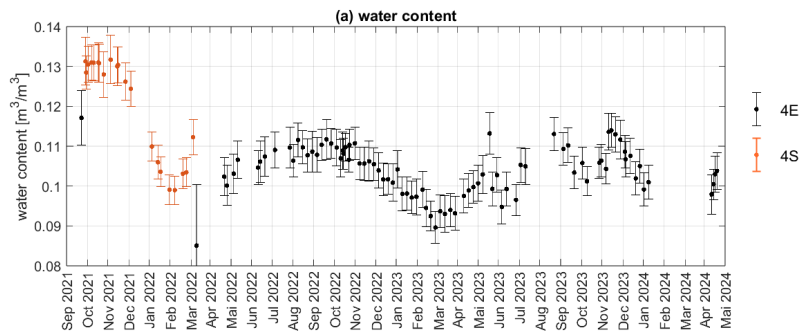
Data set 2: Ongoing measurements

- at one position
- Since Oct 2021 (with change of position in May 2022)
- Adaptation of measuring system and post-processing

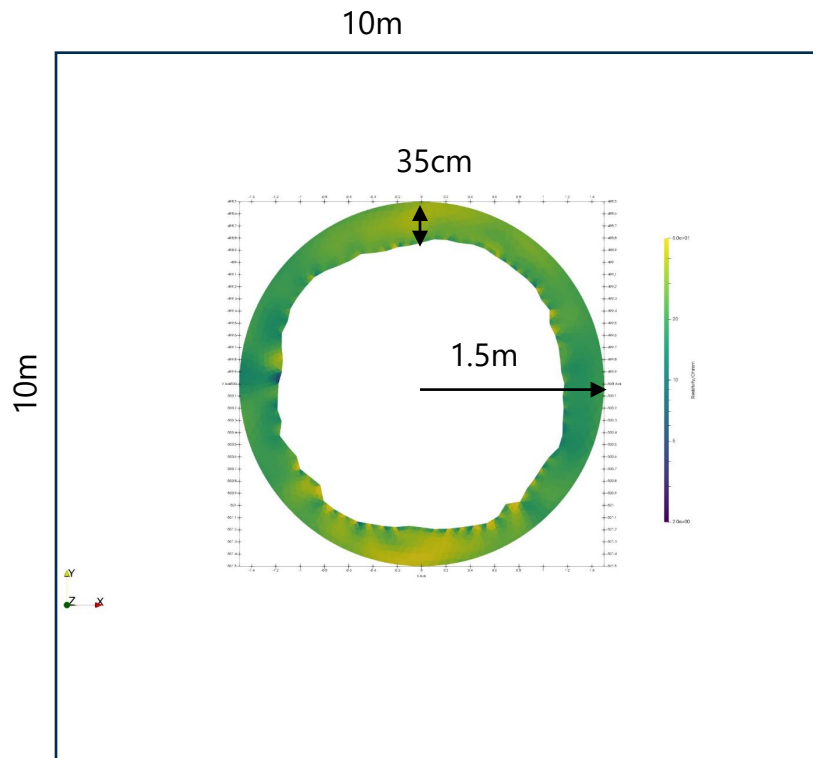
Data set 1:



Data set 2:



Available data: ERT



** I take ERT data from your archive ERT_meas_Niche_open (resistivity field).*

** Apply Archie's law to convert to water content, using the parametrization from the paper for the open twin ($\gamma = 1.44x^{-1.51}$).*

** The first measurement is 1 November 2019, which should correspond to $t = 0$ in the OGS model?*

- How do we link OGS saturation output to water content?*

$$S = WC_{vol} / n$$

with porosity:

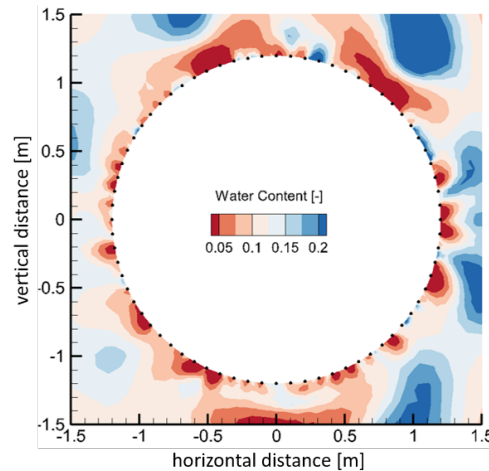
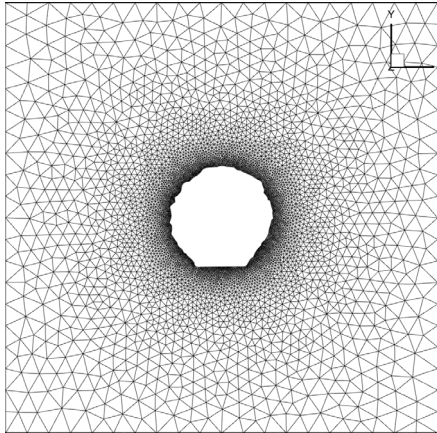
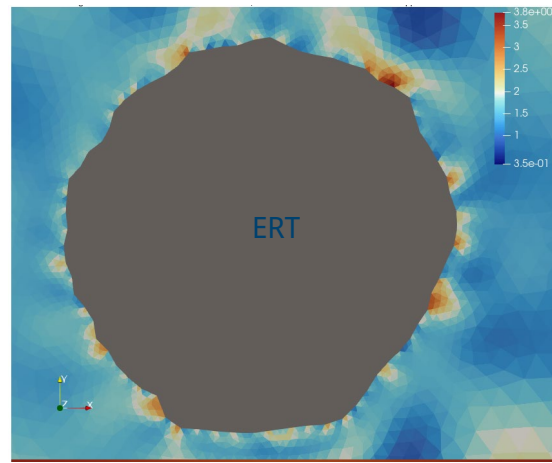
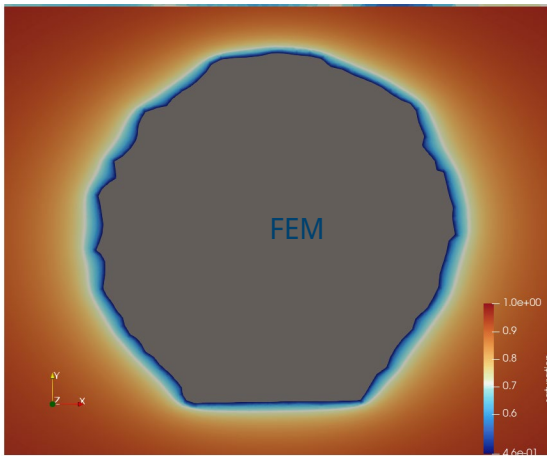
$$n = Vol_{pore} / Vol_{total}$$

** When I compare OGS output with the ERT vtk files, the mesh seems slightly different around the tunnel. Did I perhaps mixed open vs. closed twin data, or is that expected? Also, what depth of ERT measurement should I use for comparison?*

ERT



I am also attaching the screenshots of the ERT and OGS results, where the meshes around the tunnel appear different.



FEM Mesh: surface from Laser Scan data

ERT Mesh: surface with sensor positions

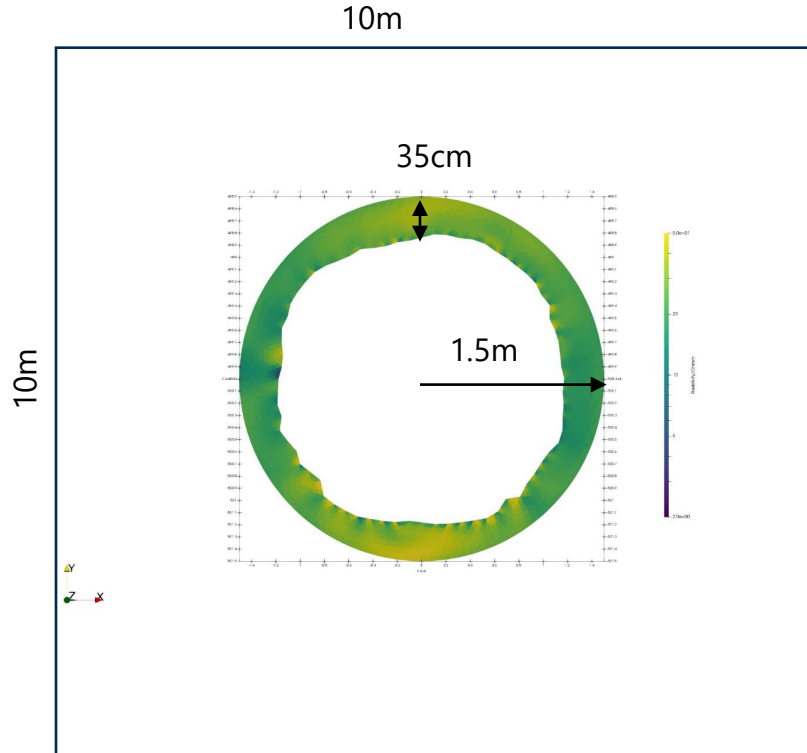
Mesh for ERT evaluation
not equal
to FEM mesh

Different surface

Different mesh

What do we have to adapt?

Available data: ERT



- Data on 10m by 10m
- Reliable range:
1.5m radius (green in the image),
35cm in the rock – only use this range!!
- FEM Definition x/y (in ERT model x/z)
- **Mesh for ERT evaluation not equal to FEM mesh**
- Data available daily (vtk)
- Data set includes several parameters (resistivity, delta resistivity, log resistivity, phase angle...)
- Data on exact circular shape

I also looked at the paper you shared, Tuanny, on heterogeneous inputs to OGS. This looks like the approach I should use. If you have example input files, that would help me adapt to this setup faster.

Heterogeneous inputs to OGS

Projection of data on elements

generate_projections.py

Input: main mesh (bulk.vtu)
Output: mesh with projections

```
meshio.write_points_cells(  
    meshOut, # str, os.PathLike,  
    points,  
    cells,  
    # file_format="vtk", # optio  
    # Optionally provide extra da  
    point_data=point_data,  
    cell_data=cell_data,  
    # field_data=field_data  
)
```

Other functions, parameters
can be defined.

*.prj

Changed to account for
projected mesh

```
cd_a_open_niche_quad.prj X  
niche_4_no_gravity_no_sigV_no_uy_deltat_projection_on_mesh > c  
1 <?xml version="1.0" encoding="ISO-8859-1"?>  
2 <OpenGeoSysProject>  
3 ....  
4 <meshes>  
5 <!--mesh>bulk.vtu</mesh-->  
6 <!--mesh>bulk_w_projections.vtu</mesh-->  
7 <!--mesh>bulk_all.vtu</mesh-->  
8 <!--mesh>cd_a_niche4.vtu</mesh-->
```

Domain identification needs
to be carried out again:

```
./bin/identifySubdomains -m  
bulk_w_projections.vtu bulk_all.vtu
```

*parameters_TRM.xml

We now read the projected
values instead of hard coded
ones:

```
03_parameters_TRM.xml 1 X  
Projekte > EURAD > EURAD_II > 100_HERMES > 003_2D_CDA  
131 <parameter>  
132 <name>phi</name>  
133 <type>MeshElement</type>  
134 <field_name>n_rd</field_name>  
135 <!--type>Constant</type>  
136 <value>0.105</value-->  
137 </parameter>  
138 <parameter>
```

